

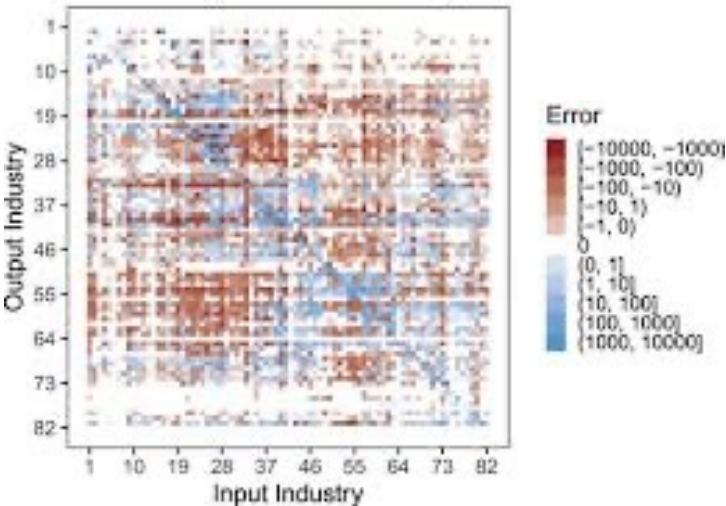
Iterative Proportional Fitting Procedure

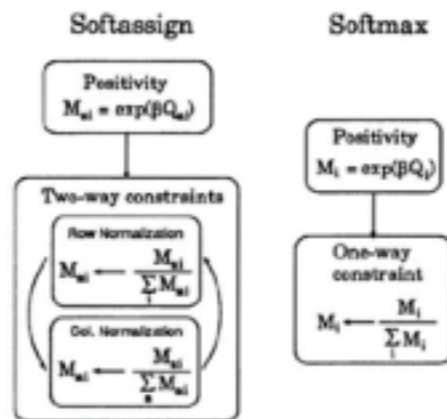
first proof of convergence

softassign

reintroduced entropy-regularized OT to machine learning

Sum of Regional Estimates by 2D RAS





Softassign versus Softmax: Benchmarks in Combinatorial Optimization

Steven Gold
Department of Computer Science
Yale University
New Haven, CT 06520-8285

Anand Rangarajan
Dept. of Diagnostic Radiology
Yale University
New Haven, CT 06520-8042

Abstract

A new technique, termed *softassign*, is applied for the first time to two classic combinatorial optimization problems, the traveling salesman problem and graph partitioning. Softassign, which

Sinkhorn Distances: Lightspeed Computation of Optimal Transport

Marco Cuturi

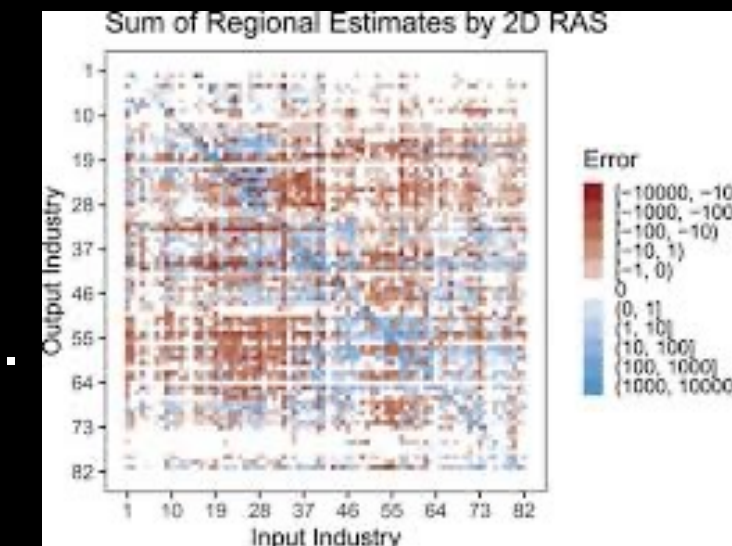
Graduate School of Informatics, Kyoto University

`mcuturi@i.kyoto-u.ac.jp`

Sinkhorn: Historical Perspective

- **Early origins**

- Iterative proportional scaling ideas go back to Yule (1912) and Kruithof (1937).
- Formalized as **Iterative Proportional Fitting Procedure** (IPFP) by Deming & Stephan (1940).
- Known in economics / OR as the RAS method (Bacharach, 1965).



- **Sinkhorn's contribution**

- Sinkhorn (1964): **first proof of convergence** for matrix scaling iterations.
- Algorithm subsequently named after him.
- Rüschendorf (1995): extension to infinite-dimensional settings.

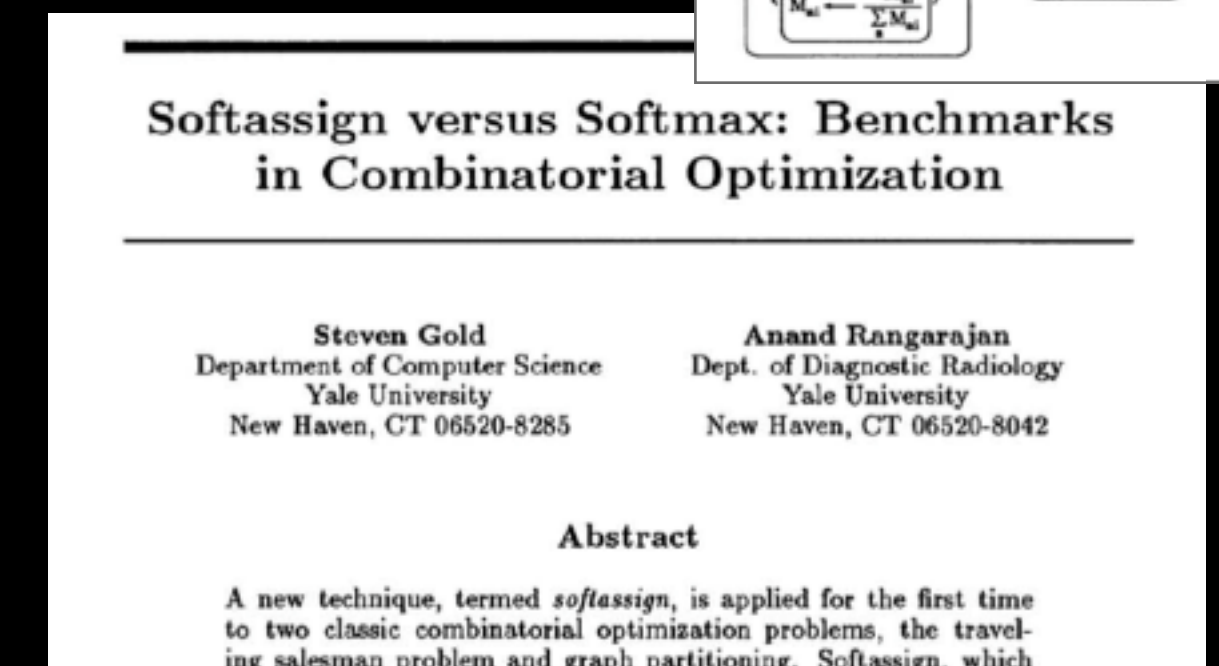
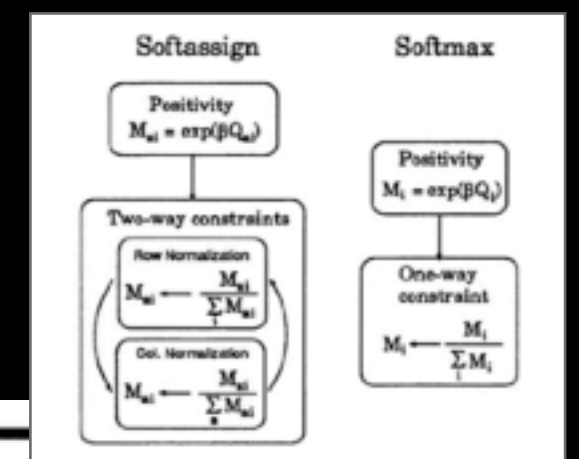
- **Adoption in economics & statistics**

- Used in Econ to approximate OT under the name *gravity models* (Wilson, 1969; Erlander, 1980).
- Rebranded as “**softassign**” (Kosowsky & Yuille, 1994) in assignment problems.
- Applied to matching in economics (Galichon & Salanié, 2009), even in ML!

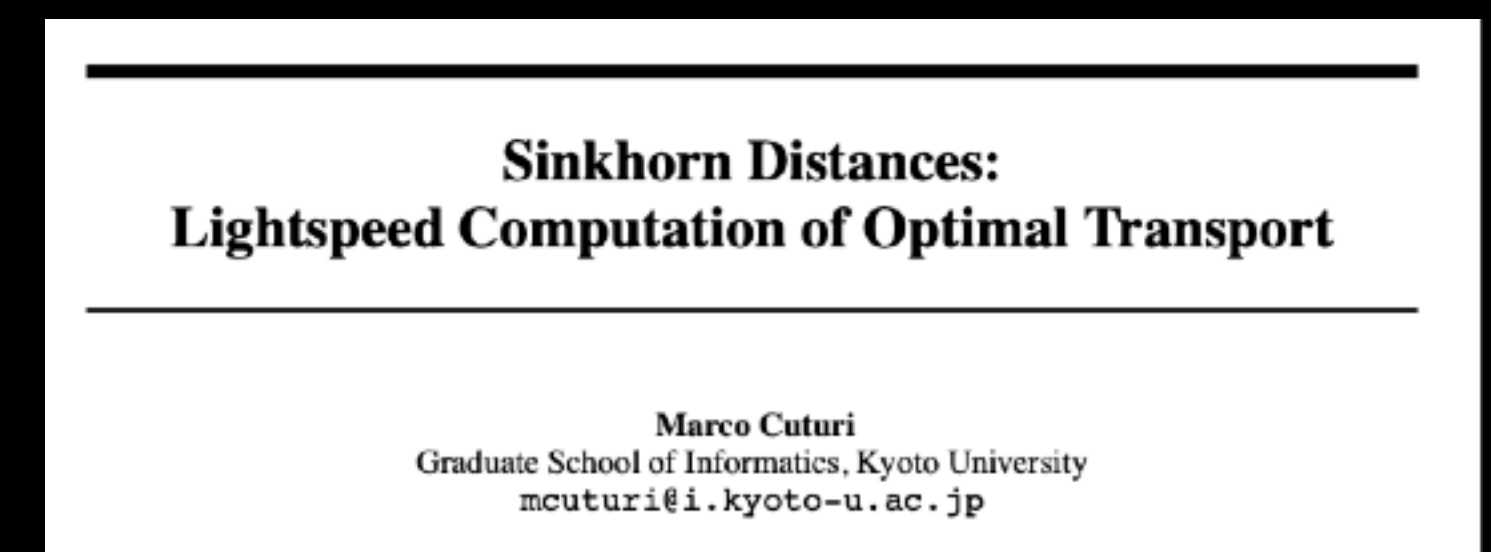
- **Modern revival / ML application:**

- Cuturi (2013): **reintroduced entropy-regularized OT to machine learning**.
 - Showed algorithm is efficient, scalable, parallelizable (GPU-friendly).
 - Produced a differentiable loss → opened door to deep learning applications.
- Countless extensions since (symmetry preservation, stochastic versions, etc.).

Holý & Šafr 2020



Gold & Rangarajan, NeurIPS 1995



Implications of EOT